

CSULB – College of Engineering
Computer Engineering
Senior Design Project
CECS490B

Roadies

Rylee Bessler
Luis Chavez
Ricardo Gallegos
Benson To
Minhhanh Tran

Prof:
Dan Cregg

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Team Roles and Responsibilities

Roles

Hardware Engineers: Minhhanh Tran & Luis Chavez

Software Engineers: Rylee Bessler & Benson To

Mechanical Engineer: Ricardo Gallegos

Responsibilities

Rylee Bessler: Safety, RP 2D LiDAR, Companion Application, Jira

At the beginning of semester A I spent the first two weeks researching safety and setting guidelines for the best approach when designing our subsystems. I created a couple of living documents within Microsoft Teams that can serve as reference points as we begin to build out these subsystems with the expectation that all individuals will update them as they progress.

I had spent some time with our original solution of RP 2D LiDAR to assist in computer vision. However, once we began to hit a processing limitation of the Raspberry Pi and needing to offload some of the processing to a neighboring computer on a shared Wi-Fi network this solution was paused.

For the end of semester A and beginning of B, I took on the responsibility of creating the companion application that was mobile focused. The application was developed using Kotlin Multiplatform, an Android first solution with the potential for cross platform development for iOS. The application served to connect with Roadie wirelessly and allow for both manual and automated modes.

Luis Chavez: Schematic, PCB Design, & Controller Programming

At the beginning of this semester, I was tasked with creating a **controller** that will make our Roadie move. This is when we started to choose parts that will go on our controller, and I decided to use an **Arduino Nano Esp32** as the microcontroller due to its speed and features. From here more research was done and the decision for our communication protocol was to use an **NRF24L01+** which uses **SPI**. By using this communication protocol, we would successfully be able to communicate with the Roadie using our controller.

Afterwards I created a rough setup of the **controller** on a **breadboard** which caused many issues, but this allowed us to see how everything would work. I **packet sized** the **data** used to send to the Roadie, which will allow it to move or use its other features, the actuator and automatic mode. Afterwards, I programmed the main Arduino microcontroller on the Roadie, the Arduino Mega 2560, where some safety protocols were established in the code when users are using the Roadie.

Finally, I created the **PCB design** of the controller using **Altium Designer**. This was to remove any type of problem that may arise when using the controller on its own as it was made on a breadboard which has a multitude of wires everywhere. This **PCB** allowed us to **compact** the controller and still has the necessary features on it.

Ricardo Gallegos: Chassis, Motor, and Scissor Jack Research and Development:

At the start of the semester, I realized I made an error with the **chassis** and had to make a new one from scratch. During the development of the second group of **dc motors**, we had **my6812 dc motors** that burned up our motor driver. The motors and driver were replaced. We needed more torque and less rpm. So, we decided to replace those motors with **worm gear motors** and have not had any issues since. The **epoxy** used to hold the axle and wheel together

ripped apart during testing for the first demonstration. This was fixed by adding a **hub** that is **welded** to the axle. The amount of torque that the robot has was more than the average parts available on the market.

The **couplers** on the drive shaft kept ripping apart. I drilled a hole through the coupler and drive shaft to create a **bolt lock**. This way, the motors and couplers are locked into place. The amount of torque needed to push the tread tracks was massive and had a potential of burning out the second motor driver if the robot was in operation for a long time.

The **tread tracks** had pvc pipe that was too large causing the robot to vibrate too much

The pvc pipe was replaced with **smaller pipes** but it made the robot move really slow.

The treads needed a way for the robot to gain traction so new sprockets were made that were smaller to accommodate the **new treads**. The new **sprockets** on the wheels kept breaking off. I found a solution to get rid of the tread tracks all-together and still have a pivoting motion like the treads.

The tread tracks were switched out with **multi-directional wheels**. These wheels allow the robot to maneuver easily and be capable of going over cables and door stops. Going over objects had the potential for the cargo to fall off the platform. The cargo that goes on top needed straps to keep the boxes in place.

The **IR sensors** were too sensitive to light, so they were replaced with ultrasonic sensors. The ultra sonic sensors then had an issue where they would cause too much delay. We switched back to the IR sensors. The controller needed a 3D printed case and an object detection camera.

The robot was painted using furniture, **wood paint and aerosol paint**. The **new scissor jack** needed a way to hold the actuator. I made a scissor jack that is roughly around the same length as the actuator in the down position. And I changed the scissor jack design from having a higher lift due to lateral force instabilities.

Minhhanh Tran: Schematic & PCB Design

This semester I have been working on a couple things to further the progress of this project. I started out researching the **RP 2D Lidar** and was able to set up the environment for that. I have since given Rylee the device so that he could do further research. After further research and limitations, we decided not to use Lidar in our project and decided to use IR sensors and the object detection camera

I then began to work on **creating a block diagram** for the overall system. This was done so that our team members, our classmates as well as myself could understand the flow of the system. I consider this block diagram to still be a work in progress that may change throughout the project; if components need to be swapped out or new measures need to be added, it can easily be done. After I created the block diagram, I used that as a reference to create the **schematic**. To go from schematic to **PCB**, I imported the footprints of each component from SnapEDA and created the layout. Each of these steps has been detailed in the Design portion of this report. The purpose of creating this power PCB was so we would not have a mess of wires underneath the Roadie, and components could be easily plugged into power. The Arduino Mega, Raspberry Pi and Oak-D Lite Camera all received 5V from the usb-c outputs, the IR Sensors received either 3.3V or 5V from the board, and the relay module and motor driver received 12V.

Benson To: Oak D Lite (object detection) and IR Sensors(Obstacle Avoidance)

Since the start of 2nd semester, after further testing ROS 2 connection between two host computers (Raspberry Pi and Laptop) which were both running on Ubuntu, I ran into troubles

with testing Wi-Fi connections between the two which led me to drop the use of ROS 2.

I have now begun work on object detection and seeing how the GPIO pins worked on the Raspberry Pi 4 in which I have started to make use of Thonny as my IDE to control the pins for the IR sensors. I will also be testing object detection on our Prodie which will help speed up development.

We made the decision to have two hardware devices for the Roadie, where a Raspberry Pi 4 would be used for IR sensors and the object detection camera which we will UART to the Arduino Mega, so we need to get a level shifter due to voltage differences in the GPIO. Due to the delay that was included through UART between the Raspberry Pi 4 and the Arduino Mega for our IR sensors, we moved the sensors from the RPi to the Mega which helped with delay.

We also ran into a problem with the IR sensors being sensitive to light which we tried to pivot to ultrasonic sensors which wasn't successful switch due to the delay that was introduced when we triggered each individual sensor at a time compared to triggering each one at the same time which could lead to incorrect detection. Due to this discovery we decided to switch back to IR sensors and try to solve the issues we were having with those sensors.

I then tested the camera with the Roadie to see what tolerance we would need when receiving the spatial data of the person when following so that we can figure out what direction the Roadie should move. This is something I would also test later as well once we removed the tracks from the Roadie due to the turn speed being faster than what the camera could keep up with due to the delay.

Project Details

Objectives:

Our Roadie project is designed to assist people in easily moving items from one location to another. Then, the addition of the scissor jack to the Roadie allows the cargo platform to rise in height allowing the user to easily place items on the platform without doing too much labor. There are many use case scenarios that the Roadie can be used such as taking items from your car to your beach spot or items from the car to the backyard. This innovation is a powerful solution that can assist anyone move items from one place to another, especially the elderly/disabled people.

Benefits to the User:

- It can easily transport heavy items from one location to another.
- Can traverse most terrains as it uses rubber tank tracks for movement.

Product Features:

- Rechargeable batteries so running out of power will not be too much of an issue.
- Scissor jack allows the carrying platform to ascend.
- IR sensors are used for obstacle avoidance.
- Oak D-Lite camera to detect the user.
- Emergency stop button in case something happens to go wrong with the autonomous forklift.



Figure 1. a) Oak D-Lite b) Raspberry Pi 4



Figure 2. a) All terrain wheels b) 6355 Motor c) Motor ESC

Design

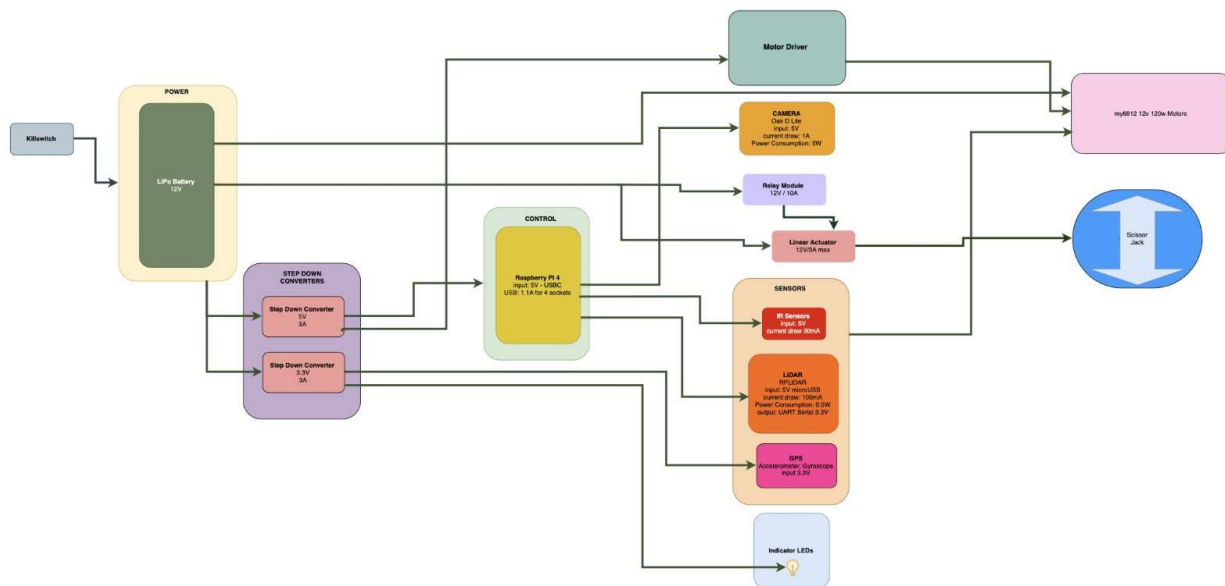


Figure 3. Block Diagram of the Roadie

Power: We will be using rechargeable 12V lithium polymer batteries to power our system and our main motor. The 12V batteries will be used to power our motors, sensors, control systems, motor driver and camera.

Motors: We will be using MY6812 12v 120W DC motors. These motors can propel the Roadie over various terrains like dirt, grass, cement, stairs, sand, and snow. The motors produce enough

torque to allow the Roadie to go up and down stairs. The Roadie will have no problems exploring rough and uneven terrain thanks to the 120w 1.2 newton-meters of force produced by the motors.

Sensors: The IR sensors will be used for object avoidance and ensure that it does not crash into anything

Camera: The camera will be used to for object detection and will be sending data to the controllers from the Raspberry Pi 4 to the Arduino Mega through UART

Control: The Raspberry Pi will be used to interact with the camera, while the Arduino Mega will be connected to the IR sensors and motors.

Killswitch: We will be implementing a kill switch. If there are any errors in the functioning of the Roadie, we will be able to kill all power to the system.

Actuator: This 12V/3A 8” actuator will be used to move the scissor jack up and down. The actuator will be directly powered by the 12V LiPo Battery and connected to the Relay Module

DPDT Relay Module- This double pole double throw relay module will be used to control the actuator. It takes 12V and a maximum of 10A input.

DC5 Dual Channel H Bridge Driver – this is an upgraded l298n that ensures the Roadie has stable and reliable power. It features a high-speed switch, suitable for various applications, the ability to handle high current loads, and can suddenly stop motors.

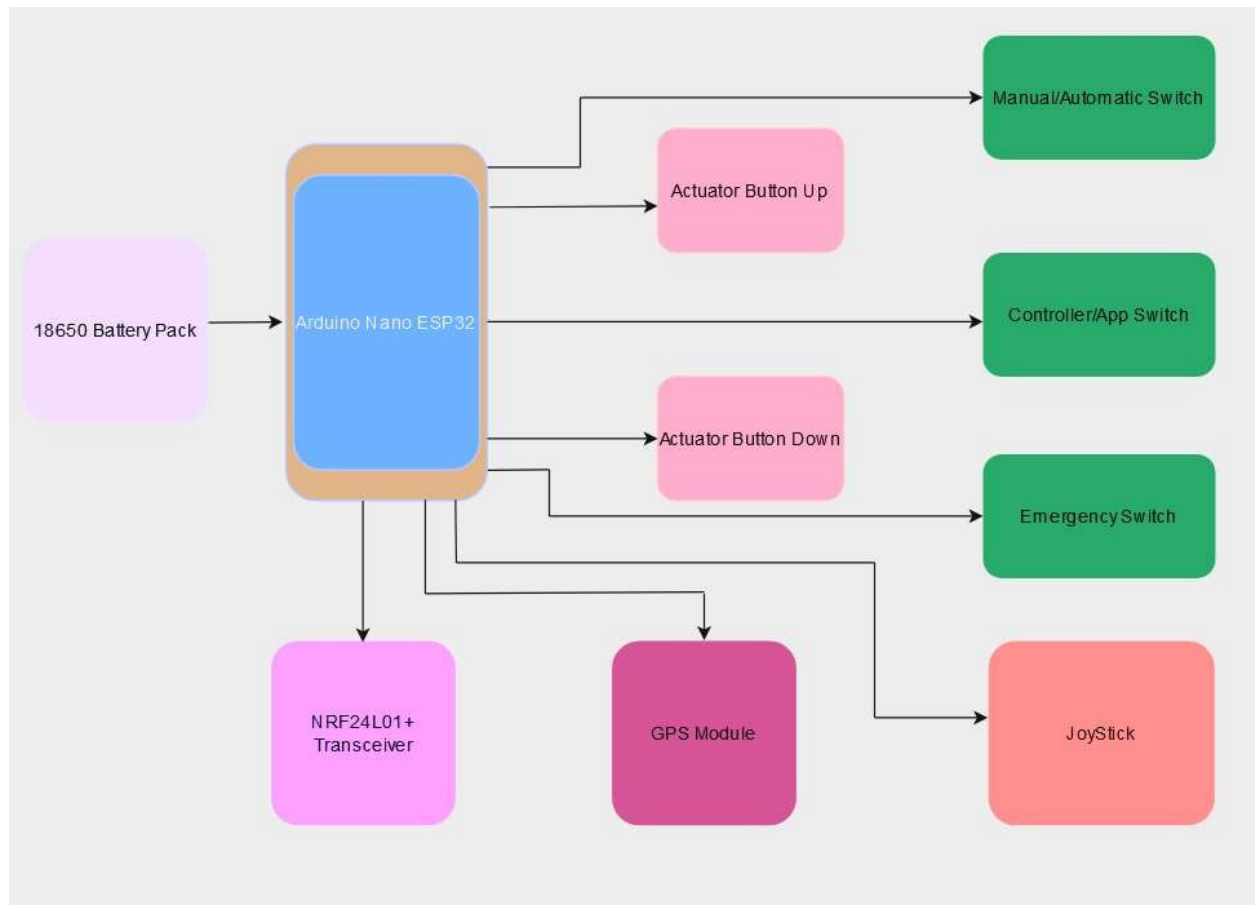


Figure 4. Block Diagram of the Controller

Power: We will be using 2 18650 LiPo batteries that will be in a battery holder to power the microcontroller via USB-C.

Microcontroller: We will be using the Arduino Nano ESP32 for the microcontroller that will be in the controller. This microcontroller is very fast, and it should be able to accomplish everything we want it for.

Joystick: We will use a dual axis joystick that considers the x and y coordinate when moving it along its axis. The joystick requires a 3.3V-5V input but we decided to use 3.3V as it is much more effective when dealing with the resolution of the ADC converter.

Transceiver: We will be using the NRF24L01+ transceiver due to its effective range of communication and will allow us to use peer-to-peer communication, which is exactly what we need to have the Roadie receive the data from the controller. The transceiver requires a 3.3V input to power on.

GPS Module: We decided to use the NEO-6M GPS module as it is very accurate for the price that it is available for and will allow us to pinpoint the position of the user. This is powered by a 3.3V-5V input but we decided to use a 5V input.

Buttons: The buttons that we will be using are the COM-10302, which are a normal set of push buttons but are big enough and have covers to make it easier for the user.

Switches: The switches we will use are the EG1248 and we went with these because they are bigger than the average switches that are out there and will allow for ease of use compared to the average switches available.

Schematics:

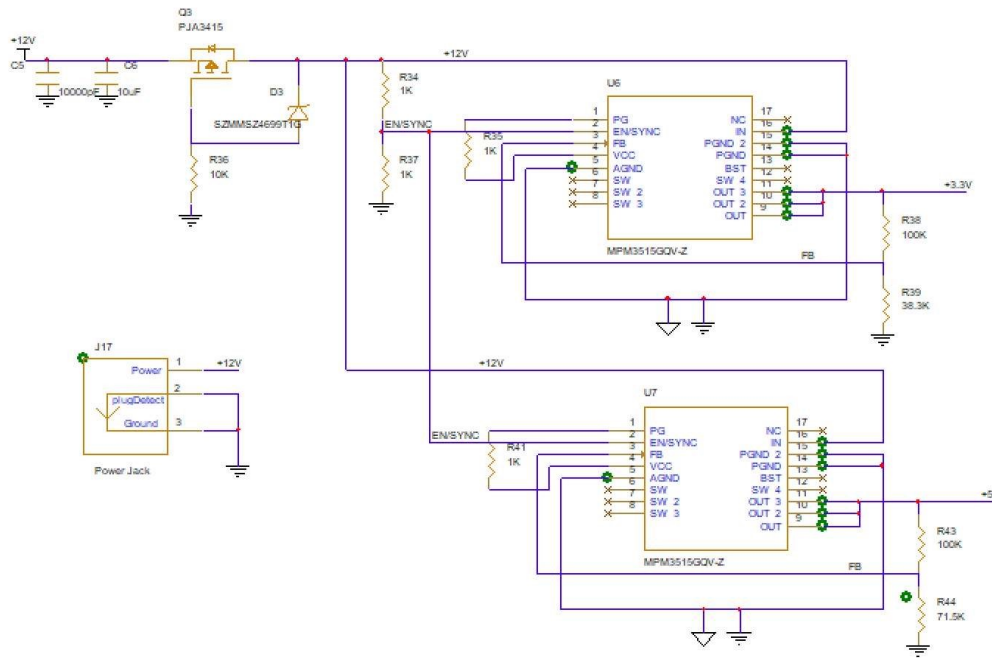


Figure 5. Input - Schematic

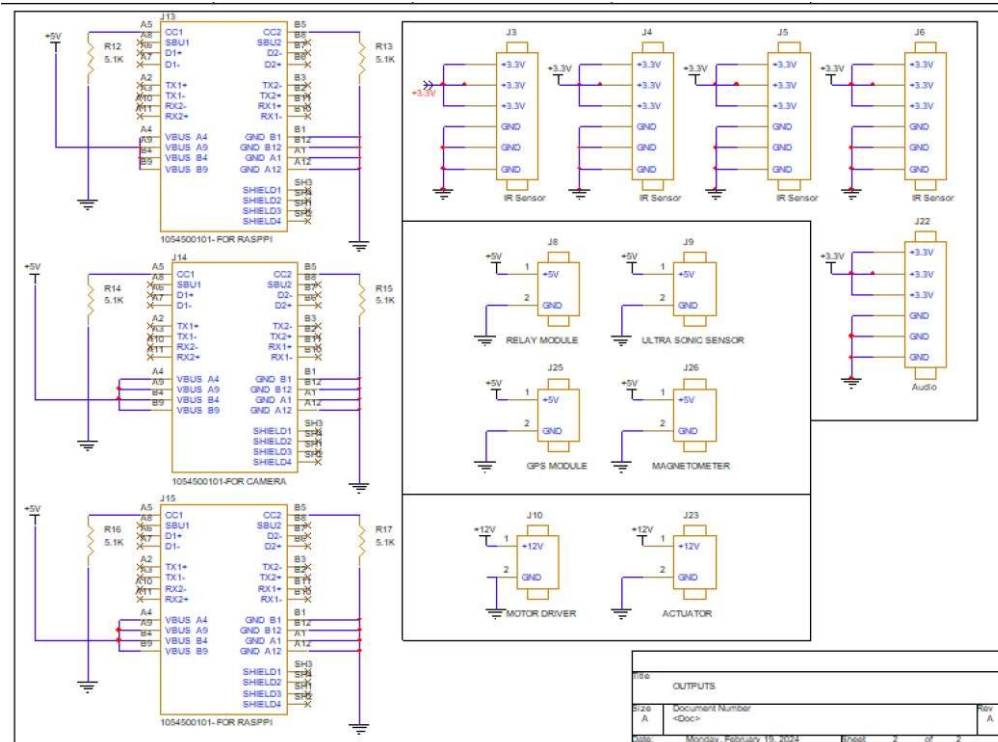


Figure 6. Outputs - Schematic

PCB Component Placement:

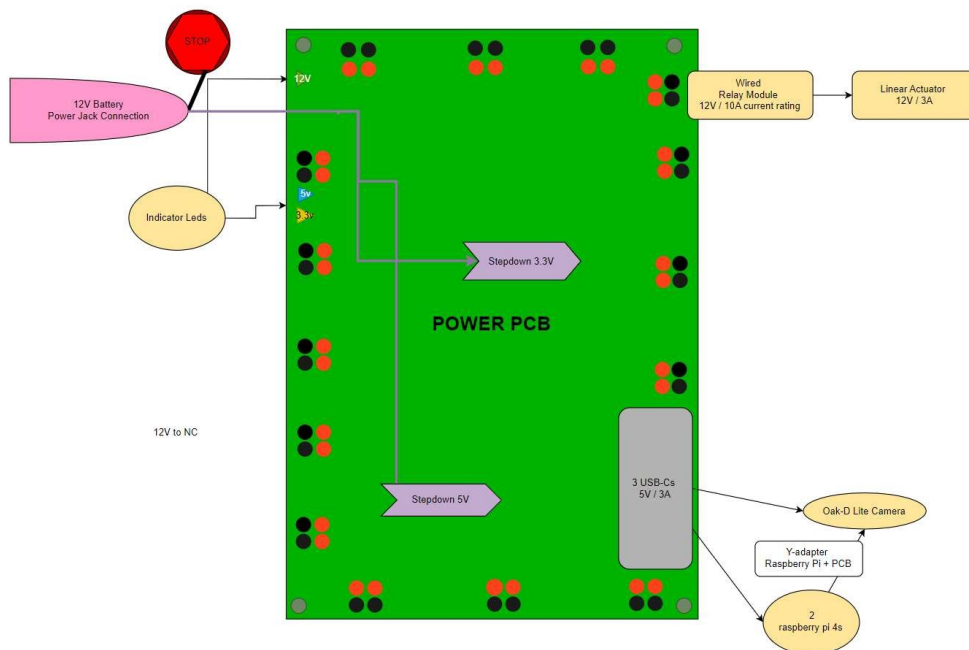


Figure 7. Rough Draft of where parts will be placed on the PCB

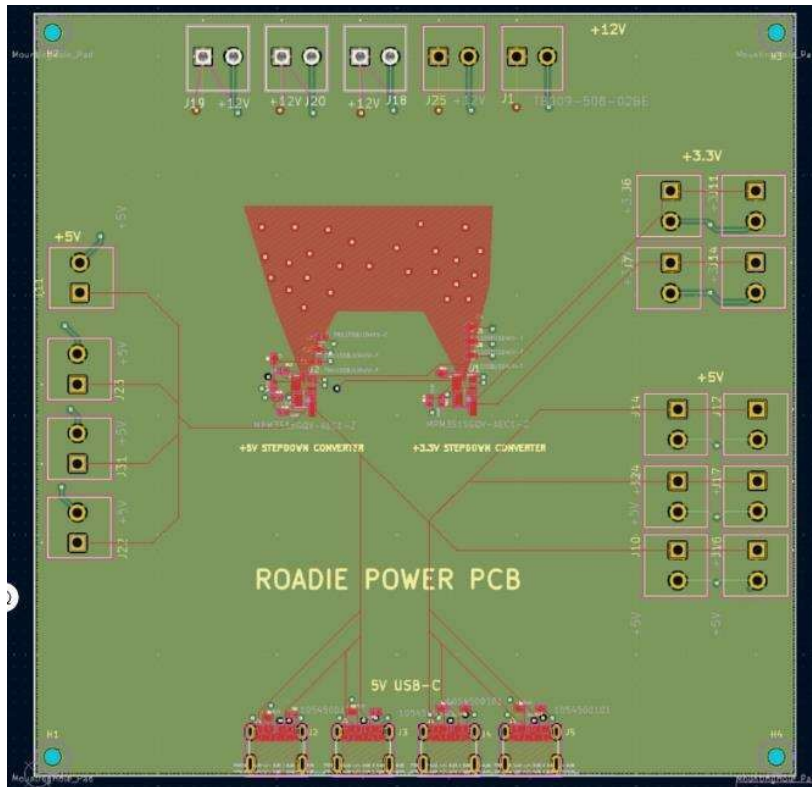


Figure 8. Final PCB Layout

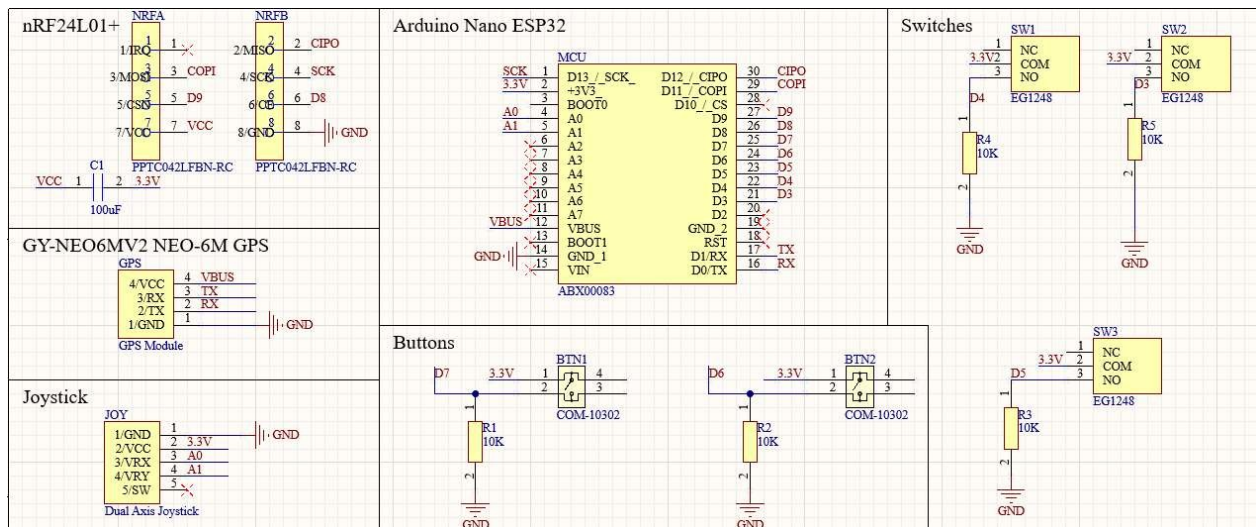


Figure 9. Controller Schematic

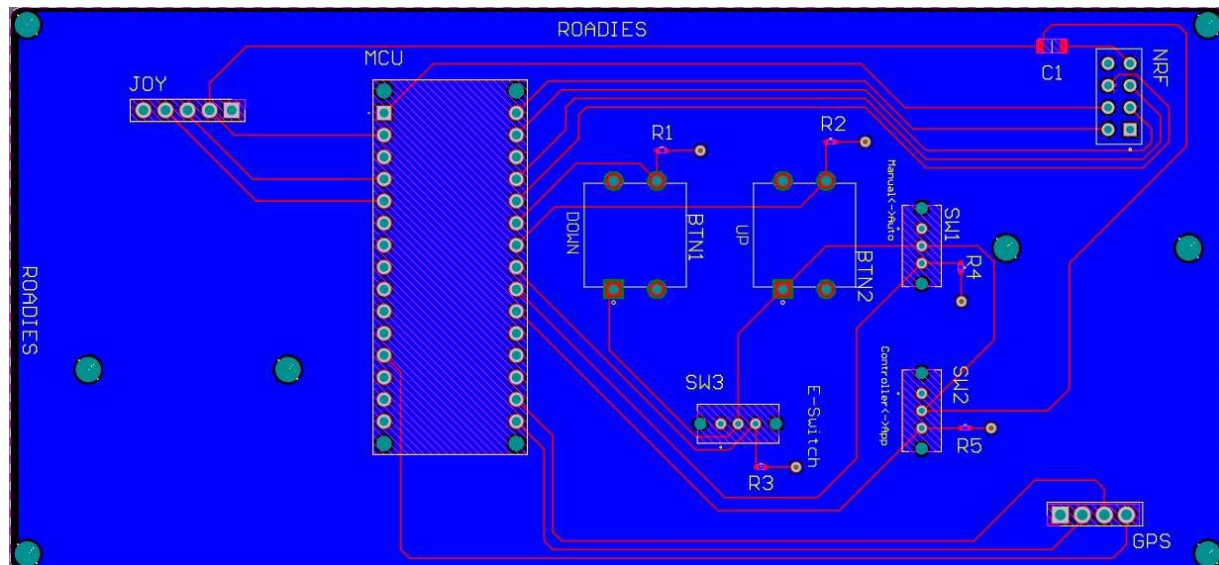


Figure 10. Controller PCB

Subsystem interfacing:

IR Sensors & Oak D lite:

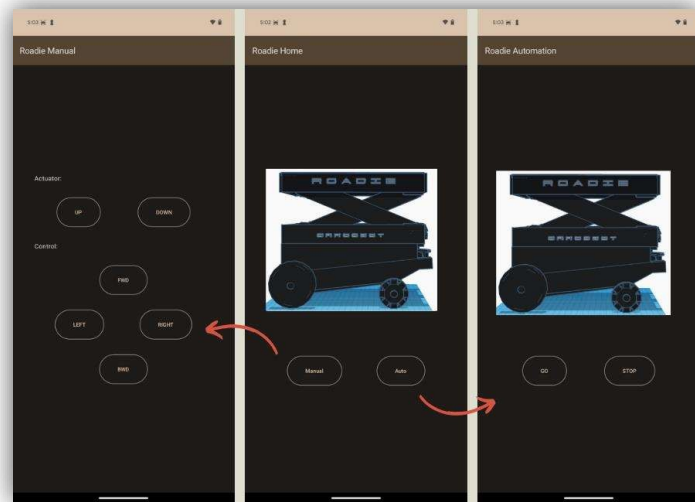
The physical interface for the Oak D lite is USB3(Universal Serial Bus) while the IR sensors will be connected to GPIO pins on the Arduino Mega.

The Roadie will take in data from the IR sensors while in autonomous mode, which will interface with the motors and determine what directions they should move depending on which IR sensors are detecting an object. Once the Roadie has detected the person it is trying to follow, the Oak D Lite (object detection) will track the person while controlling the motors by sending data over to the Arduino Mega over serial UART from the Raspberry Pi 4 through the use of a level shifter as there is a voltage different between the two hardware, but the IR sensors will have a higher priority to the detection camera for safety purposes.

Companion Application

Early iterations of the mobile application were created using html websites on the mobile phone side that would communicate with the Raspberry Pi via a Wi-Fi connection using Flask and UTF-8 encoding to translate to Python code for the Raspberry Pi to interpret. This version of the application was made possible through Ricardo's efforts.

The companion application was developed on Kotlin's Multiplatform Beta for parallel development with both Android and iOS. With the multiplatform beta iOS development was still requiring additional work to ensure full functionality and often had aspects of the code base 'break' upon update. This along with the knowledge that iOS would also require additional hardware to UART via Bluetooth to Roadie being that it was incompatible with our already owned HC-05s lead to the abandonment of the iOS portion of the application.



To elevate the companion application's functionality NavHost was used to create a multiscreen layout to navigate to the different modes of Roadie; Manual mode and Automatic mode. With this version of the application Bluetooth permissions were granted in the xml file with MainActivity facilitating the connection to Roadie. However, this solution also required a Bluetooth socket to be passed through three separate screens in the application's startup which led to a buffer issue with the data being sent to the HC-05 which was misinterpreted on the tested Arduino One. This resulted in the need to create a ViewModel which would handle Bluetooth communication to streamline the process. Following the creation of the ViewModel, the application would successfully connect to Roadie, the application would halt on the initial draw of the homescreen. This result was not successfully triaged with the time left in the semester before final demos.

Power Systems:

We are planning to only use one power source to operate the Roadie. The 12V rechargeable lithium battery. It has a maximum continuous current of 100A and max continuous load of 1280W. This battery will be powering components on the board such as the Raspberry Pi, camera, and lidar which all need a 5V input, the LEDs and the Digital IMU which need a 3.3V input, as well as the Nema 17 motors which will be used for the 360° arms. We have ensured the needed voltage and current ratings are within the power sources' requirements.



Figure 11. 12V LiPo Battery

CPU processing requirements, core and/or per subsystem:

There were multiple discussions as to what microcontroller will fit our needs for this project. It was narrowed down to an NVIDIA Jetson and the Raspberry Pi 4. We concluded that the Raspberry Pi 4 will be the much better option to go for as it is cheaper and will have similar use cases to the Jetson. The specifications of the Raspberry Pi 4 that we are using are: Quad core Cortex-A72(ARM v8) 64-bit SoC @ 1.8GHz, 4gb of LPDDR4-3200 SDRAM, 2.4 GHz and 5.0 GHz IEEE 802.11ac wireless, Bluetooth 5.0, BLE Gigabit Ethernet, 40 pin GPIO headers, 2 USB 3.0 ports, 2 USB 2.0 ports, 2-lane MIPI DSI display port, and 2-lane MIPI CSI camera port. The laptop we are currently using to test the usage of Nav2/ROS 2 with the Raspberry Pi has an AMD Ryzen 5 5500U with Radeon Graphics which has 6 cores and 12 threads, with 6GB of DDR4 3200 MHz memory, and 2GB for vram.

Electro-Mechanical:

The Roadie features several mechanical requirements to ensure its functionality and versatility. The Roadie is equipped with two powerful 36v BLDC motors, offering plenty of torque for both forward and backward movement. With a continuous track system, the Roadie is designed to handle various terrain. The maneuverability features four 10-inch wheels that allow the Roadie to maneuver through different environments with ease. Each of the Roadie's wheels is reinforced with a robust ½" axle, ensuring it can reliably support test loads of up to 170lbs so far.

To maintain cargo bed stabilization, a 12v linear actuator is integrated into the rear frame, which reacts to offset movements detected by the gyroscope, automatically centering them. A continuous track system is made up of 28 1" pvc pipes cut to 4" on each track for durability. The Roadie's bike chain requires the removal of four links on each side for proper operation. The chassis incorporates 1-inch holes every 2 inches to reduce overall weight, enhancing efficiency and mobility. Another 12v linear actuator is included to manage the lowering and raising of the scissor jack for added functionality. The Roadie is well equipped to handle a wide range of tasks and environments while maintaining cargo stability. To enhance the durability and load-bearing capacity of our continuous track structure, we upgraded the PVC pipe diameter to 1 inch. The new PVC pipe can withstand pressures up to 450lbs before showing deformation.

I encountered challenges with the spacing between the bike chains in the second set of PVC pipes. The initial configuration prevented the wheels from resting on the PVC pipes, blocking their rotation. This required me to re do the whole tread build and purchase more PVC pipe. After several adjustments, there was a precise fit between the treads and wheels ensuring optimal performance. A DIY spring or suspension is being considered in the future to distribute the weight onto the frame struts rather than relying on the motor axle. Drawing from lessons learned, I recognized the importance of mitigating long term risks associated with the motor axles bending under heavy loads. A solution is mounting the motors to the chassis coupled with the use of a ¼" to ½" axle rod adapter. The ½" axle rod will be supported by a pillow block bearing, effectively dispensing the weight evenly onto the ½" axle. Reducing the strain on the motor axles and ensuring the Roadie's long-term reliability and performance. I chose 6355 270kv BLDC motors, which will be energized by a 36v 4ah 10s2p Lithium-ion battery managed by a 10s version speed controller. These motors can generate up to 13lbs/ft of torque, ensuring the Roadie's capability to handle various tasks easily. The Roadie will also be equipped with 12v linear actuators, each with 3Ah capacity and the ability to support up to 330lbs of weight. These

specifications far exceed the anticipated stresses the Roadie will encounter, as our cargo weight is limited to 42lbs. This ensures that we never reach the limits on any of our components.

The Roadie's overall system weight is 18 lbs and is 24 inches in length and 18 inches in width. While the exact height is still pending, I anticipate it to be approximately 18 inches. Fuse circuits will be implemented to safeguard the electrical components from potential overloads. An Audible circuit breaker will be integrated to alert users in case of unforeseen issues. The Roadie features micro switches serving as emergency stop kill switches. These switches require manual resetting before the Roadie can resume its operation, effectively proving a failsafe mechanism in case of emergencies. We may also include speed sensors to ensure that the Roadie remains within the prescribed speed limit, which is capped at 4mph. As for our current stage of development, the Roadie is equipped with a dual electric speed controller, necessitating XT60 battery series adapters with 14-gauge silicone wire for power connectivity. Soon, we anticipate additional circuit boards, and we acquire the necessary batteries.

This semester, to make things easier, I decided to go with MY6812 motors to replace the previous motors. The VESC was replaced with an updated version of the L298n motor driver. This driver and new motors run at 12v eliminating a secondary power source that would have been needed to power the old motors. The dimensions of the Roadie were changed from 24"x18" to 24"x24". These new dimensions still allow the Roadie to squeeze through door entrances easily. The extra added space was due to the placement of the new motors. The new motors are 4.2" in length and are set up next to each other. For the motors to sit flat side by side, the Roadie had to be made slightly wider to accommodate the new motors. Scooter wheels of 100mm (about 3.94 in) in diameter were added to replace the front wheel. This allows the Roadie to be all-terrain. This means that the Roadie will have a 45-degree continuous track front end that will allow the Roadie to naturally climb up any stairs that are less than 9.5" in height.

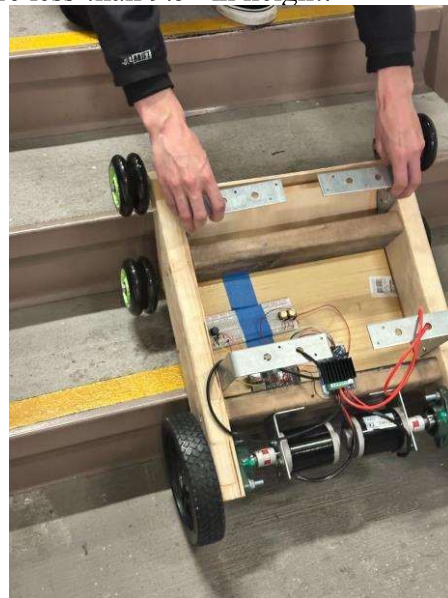


Figure 12. Updated Chassis of Roadie

Verification

Prior to proceeding with additional tasks, I conducted a weight stress test to assess the Roadie's chassis stability. The objective was to determine its capability to support one case of water bottles, and then 2 cases. The initial test involved placing a single case of water bottles on the Roadie, which weighed about 32 pounds. The Roadie performed great displaying no signs of difficulty to move. We increased the weight by adding a second case of water bottles, raising the total weight supported by the Roadie to 95 pounds. Despite the increased load the Roadie continued to function effectively demonstrating its ability to handle the additional weight.



Figure 13. The Roadie chassis and first water bottle case test weighing 36.4lbs.



Figure 14. A case of water bottles weighs 36.4lbs



Figure 15. a) The total system weight of the Roadie is 18 lbs b) The Roadie is successfully able to move effortlessly with 1 case of water bottles.

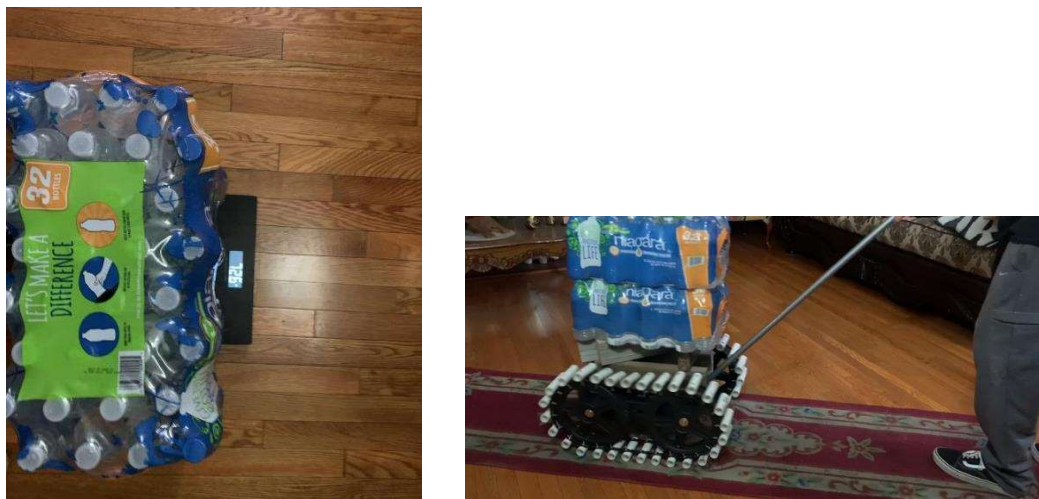


Figure 16. The Roadie can move the cases of water bottles (weighing 72.6lb), but the cross members began to camber out due to the excessive weight.

The screws used to fasten the cross member to the frame's length displayed a noticeable deflection curving outward like negative camber during the weight stress test. Upon removing the applied weight, the temporary offset camber on the continuous track would promptly revert to its original position. To prove a visual representation of this, we have included the following graph, illustrating the results of the weight stress test.

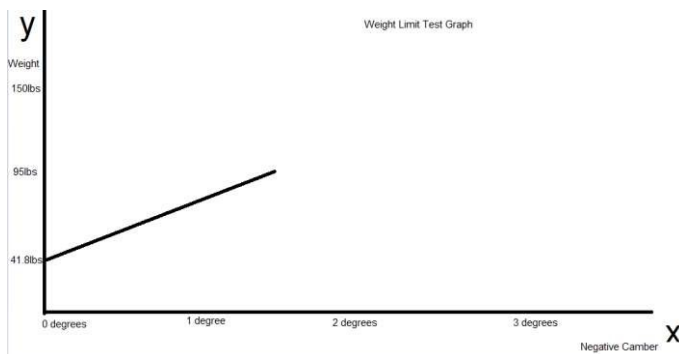


Figure 17. Results of weight stress testing

Ensuring the Roadie's robust capacity to bear substantial weight is very important. The capability is crucial for the incoming inclusion of components such as batteries and a scissor jack, which are expected to introduce a significant increase in overall weight. As of now, without any additional weight or components, the Roadie weighs 18 pounds but will be lighter and stronger with an aluminum chassis.

Tools and Methods The tools we have used so far are listed below:

Team/ Organization:

- Microsoft Teams
- Jira Timeline

Hardware:

- OrCad- for schematic design
- Draw.io - for block diagram
- Falstad.com - for simulating circuits
- JLCPCB - for fabricating the board
- Soldering Iron - for assembling the board

Software:

- Raspbian - OS installed on Raspberry Pi V12 Bookworm
 - o IDE - Thonny
- Kotlin Multiplatform Beta
 - o IDEs - Android Studio, Xcode, Arduino IDE
 - o Mobile Devices were running Android 14 and iOS17

Mechanical:

- Arc Welding machine - for future aluminum chassis fabrication
- Benchtop drill press - to accurately drill holes into wood and metal
- Abrasive metal chop Saw - to cut metal bars accurately
- Metabo HPT Miter Saw - used for cutting wood in a straight or angled line
- Screws & nuts - to hold the bicycle chain together to the pvc pipe
- Self-centering dowel jig - aligns your drill spot accurately so that you can drill holes perfectly centered in the middle of a metal bar or wood plank.
- Right angle frame clamps - allow you to join 2 pieces of wood by holding them perfectly at 90 degrees.
- 3D printer - used to print out spacers that allow more surface area to glue epoxy to the wheels.
- Epoxy glue - 2-part adhesive that forms when you mix epoxy resin and hardener.
- Tape measure - flexible ruler used to measure length
- Leveler - device for establishing a horizontal plane
- Protractor - used to draw perfect circles on plywood
- Triangle Ruler - an object used in engineering that provides a straightedge at a right angle.

Technical Data



Figure 18. Prototype 1 of the Roadie's Scissor Jack – both cad and actual design

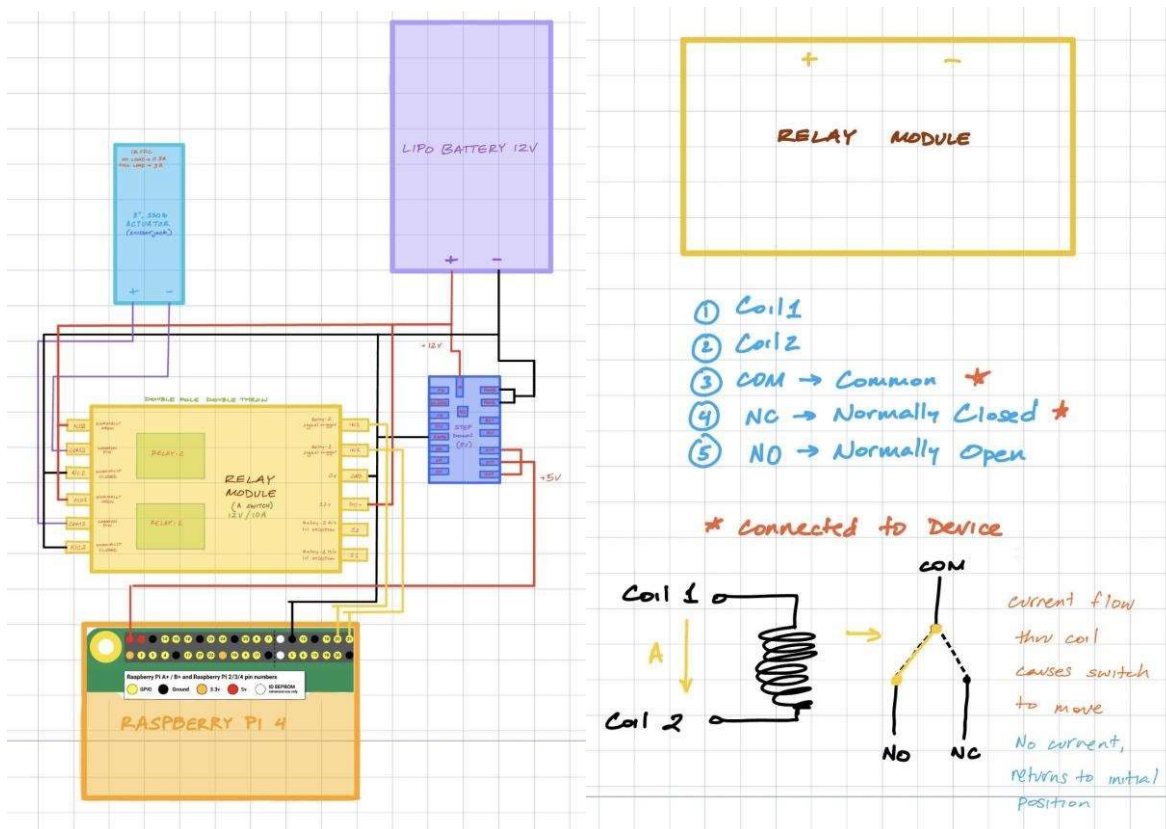


Figure 19. Block Diagram of Scissor Jack Design & DPDT Relay Module

```

1  #include <SPI.h>
2  #include <RF24.h>
3  #include <printf.h>
4  #include <stdio.h>
5  #include <stdlib.h>
6
7  // Pins for the NRF
8  #define MISO_PIN 12
9  #define MOSI_PIN 11
10 #define SCK_PIN 13
11 #define CE_PIN 8
12 #define CSN_PIN 9
13
14 // Variables
15 int32_t joy_x, joy_y;
16 byte top_button, bottom_button, emergency_button, auto_button;
17
18 //Creates the radio
19 RF24 radio(CE_PIN, CSN_PIN);
20
21 // Address that will be the pipeline
22 const uint64_t address = 0xE8E8F0F0E111;
23
24 // 12 Byte struct for the controller (need to add one more Byte for one more switch on the controller)
25 typedef struct Controller{
26     int32_t x;
27     int32_t y;
28     byte t_button;
29     byte b_button;
30     byte e_switch;
31     byte a_switch;
32 } Controller_t;

```

```

33
34     void grab (Controller_t* data);
35
36 > void setup() { ***
51     }
52
53     // Grabs the controller data
54 > void grab(Controller_t* data){ ***
61     }
62
63 > void loop() { ***
98     }
99
100     // TESTING SERIAL PRINTING
101
102     // Serial.print("x: ");
103     // Serial.print(reading.x); //print joystick x
104     // Serial.print("\t"); //Tab Space
105     // Serial.print("y: ");
106     // Serial.print(reading.y); //print joystick y
107     // Serial.print("\n"); //Newline
108     // Serial.print("top_button: ");
109     // Serial.print(reading.t_button); //print top button
110     // Serial.print("\t"); //Tab Space
111     // Serial.print("b_button: ");
112     // Serial.println(reading.b_button); //print bottom button
113     // Serial.print("e_button: ");
114     // Serial.println(reading.e_button); //print emergency button

```

Figure 20. Controller Transmitter Code

```

1     #include <SPI.h>
2     #include <RF24.h>
3     #include <printf.h>
4     #include <stdio.h>
5     #include <stdlib.h>
6
7
8     #define CE_PIN    30
9     #define CSN_PIN   32
10
11     //Motor Pins
12     const int in1A = 6;
13     const int in1B = 5;
14     const int in2A = 4;
15     const int in2B = 3;
16     const int enA = 7;
17     const int enB = 2;
18
19     // IR Sensor Pins
20     const int irSensor1Pin = 25; // Sensor 1 pin
21     const int irSensor2Pin = 26; // Sensor 2 pin
22     const int irSensor3Pin = 27; // Sensor 3 pin
23
24     // Switch variable
25     int movement;
26
27     //Variables
28     byte top_button, bottom_button, emergency_switch, auto_switch;
29     //creates the radio
30
31     RF24 radio(CE_PIN, CSN_PIN);
32

```

```

32
33 // 12 Byte struct for the controller (need to add one more Byte for one more switch on the controller)
34 > typedef struct Controller{ ***
41 } Controller_t;
42
43 //Address that will be the pipeline
44 const uint64_t address = 0xE8E8F0F0E11L;
45
46 Controller_t read;
47
48 > void setup() { ***
71 }
72
73 // Calculate turning motor speeds based on joystick x-axis value
74 int motor_speed_A;
75 int motor_speed_B;
76 int x_motor_speed;
77 bool Statement = false;
78 > void loop() { ***
432 }

```

Figure 21. Roadie Receiver Code

Demo List & Outcomes

Demo #1: Prototype & Motion Control

- Demonstrate the actuator and its ability to extend and collapse our scissor jack fully while respecting the jack's minimum and maximum height limitations.
- Demonstrate the motors functionality and ability to provide accurate measurements back to the user.
- Demonstrate the distance the Protie traveled and the distance that the actuator reached while supporting an item.

Outcome: We were able to show video of the Protie working in action but when trying to show a live demonstration, this is where we faltered as we were unaware about how to go about using the school Wi-Fi. The live demonstration would be considered unsuccessful as we were unable to demonstrate our prototype in action but were able to make up for it via videos so in terms of functionality it was successful.

Demo #2: Handheld Controller and Movement over a Terrain

- Demonstrate the emergency shut off system for Roadie via the emergency stop button and reboot.
- Demonstrate a controller that will drive and control the Roadie using P2P communication.
- Demonstrate Roadie's ability to traverse across harsh terrains.

Outcome: We blew a motor driver right before our live demo, but we were able to make it up by showing a video as to how the Roadie will move. The live demonstration was unsuccessful, but we were able to show the functionality success we had with the Roadie through recordings of our progress.

Demo #3: Autonomous Control (using IR, GPS, and Computer Vision)

- Demonstrate the Roadie in a fully autonomous following mode. In this mode, Roadie will continue to demonstrate any previously shown demos alongside improvements made to those systems.

Outcome: The video demonstration was a success as we were able to show the Roadie following a user and seeing how it navigates around obstacles with its sensors. This demo was successful in displaying what our objective was.

Final Demo: App Integration & Switching Modes

- Demonstrate connecting the mobile application with Roadie.
- Demonstrate the ability to control Roadie with manual controls, while Roadie remains aware of its surroundings.
- Demonstrate any automated modes enabled on Roadie.
- Demonstrate ability to raise and lower scissor jack.

Outcome: We were able to demonstrate switching modes between manual and automatic mode for the Roadie, but we were unsuccessful in creating a mobile application for the Roadie as there were issues with Bluetooth and backend errors occurring but overall, all it was successful.

Bill of Material and Cost

Cost of Parts

Purchased Items	Mfg	Supplier	Status	Quantity	Price	Total
2D LIDAR	Slamtec	Amazon	Obtained	1	\$99.00	99
OAK Y-Adapter	Luxonis	Amazon	Obtained	1	\$20.66	\$20.66
OAK-LITE OBJECT DETECTION CAMERA	Luxonis	Amazon	Obtained	1	\$189.00	189
RASPBERRY PI 2GB MODEL B	Raspberry Pi	Amazon	Obtained	1	\$70.00	70
RASPBERRY PI 4GB MODEL B	Raspberry Pi	Amazon	Obtained	1	\$66.95	66.95
RASPBERRY PI TOWER COOLER	Raspberry Pi	Amazon	Obtained	1	\$16.99	16.99
RASPBERRY PI POWER SUPPLY	Raspberry Pi	Amazon	Obtained	1	\$8.00	8
MICRO HDMI CABLE	Vilros	Amazon	Obtained	2	\$15.00	\$30.00
EMERGENCY KILL SWITCH (set of 2)	Generic	Amazon	Obtained	1	\$6.99	6.99
2 Channel Relay Module 12v	HiLetgo	Amazon	Obtained	1	\$8.59	8.59
12v 3A LINEAR ACTUATOR	Eco-Worthy	Amazon	Obtained	1	\$42.99	42.99
Arduino Nano ESP 32	Arduino	Amazon	Obtained	1	\$21.00	21
4pcs NRF24L01+ Wireless Transceiver Module	HiLetgo	Amazon	Obtained	1	\$7.89	7.89
Dual-axis XY joystick Module	Teyliten Robot Store	Amazon	Obtained	1	\$8.99	8.99
Mini Micro slide switch 1p2t 2 position 70pcs	Chanzon	Amazon	Obtained	1	\$5.99	5.99
80 Piece Female to Female Jumper Wires	GenBasic	Amazon	Obtained	1	\$6.49	7.06
Brushed Motor, MY6812 12V 120W DC Brush High Speed Motor	Vestlife	Amazon	Obtained	2	\$35.66	71.32
12 Gauge Silicone Wire 10 ft red and 10 ft Black	BNTECHGO	Amazon	Obtained	1	\$12.98	\$12.98
New Raspberry Pi 3 Model B	Raspberry Pi	Amazon	Obtained	1	\$40.00	40
80 Pcs Copper Wire Lugs, Battery Terminal Connectors	Dizzily	Amazon	Obtained	1	\$8.99	8.99
L298N Dual Motor Driver 4 pack	Bojack	Amazon	Obtained	1	\$9.99	9.99
CO-WORTHY 5Amp 12V Automatic Smart Battery Charger	Eco-Worthy	Amazon	Obtained	1	\$49.99	49.99
Raspberry Pi, Expansion Board GPIO 1 to 4	GeeeeeekPi Store	Amazon	Obtained	1	\$9.99	9.99
Anker portable charger power bank 5/2A	Anker	Amazon	Obtained	1	\$25.95	25.95
ESP32S ESP32 ESP-WROOM-32 Development Board	Teyliten Robot Store	Amazon	Obtained	4	\$16.88	67.52
DCS-12V 0A-30A Dual Channel H Bridge Driver Motor	WAlfront Store	Amazon	Obtained	1	\$26.92	26.92
4PC Dual DC 3-6V Gearbox motor	D-life	Amazon	Obtained	1	\$7.99	7.99
HiLetgo GY-NEO6MV2 Neo-6m GPS Flight Controller Module	HiLetgo	Amazon	Obtained	1	\$8.99	8.99
12V 30AH LiFePO4 Battery, Rechargeable Lithium Ion Phosphate	Eco-Worthy	Amazon	Obtained	1	\$95.99	95.99
ARC WELDING MACHINE	Dekopro	Amazon	Obtained	1	\$124.99	124.99
LEATHER WELDING GLOVES	Lincoln Electric	Amazon	Obtained	1	\$13.99	\$13.99
FLAME RESISTANT WELDING SLEEVES	Lincoln Electric	Amazon	Obtained	1	\$10.98	10.98
WELDING HELMET	Yeswelder	Amazon	Obtained	1	\$39.99	39.99
FLEXIBLE SHAFT COUPLER SET OF 2	Rigid	Amazon	Obtained	1	\$14.99	14.99
BICYCLE CHAIN	Zonkie	Amazon	Obtained	9	\$16.88	151.92
PVC PIPE 1"X5'	Apollo	Home Depot	Obtained	6	\$10.96	65.76
MACHINE SCREWS #8-32X1" 100pack	Everbilt	Home Depot	Obtained	3	\$5.99	17.97
MACHINE SCREW NUTS 100pack	Everbilt	Home Depot	Obtained	3	\$5.99	17.97
BIKE LINK PLEER & CHAIN BREAKER TOOL	Bleko	Amazon	Obtained	1	\$17.99	17.99
ABRASIVE METAL CHOP SAW	Vevor	Amazon	Obtained	1	\$111.99	111.99
BENCHTOP DRILL PRESS	Pioneer Works	Amazon	Obtained	1	\$109.99	109.99
C CLAMP LOCKING WELDING PLIERS Spack	YaeGarden	Amazon	Obtained	1	\$32.92	32.92
PUSH MOWER WHEEL KIT (set of 4)	Parts Camp Store	Amazon	Obtained	1	\$34.99	34.99
Metabo HPT Miter Saw	Metabo	Amazon	Obtained	1	\$119	119
Pillow block bearings 1/2" bore 4pack	PGN	Amazon	Obtained	1	\$31.60	31.6
8mm to 10mm Bore Rigid Coupling Set Screw	uxCell	Amazon	Obtained	1	\$7.49	7.49
Scooter Wheels 100mm (4pack)	FreeDare	Amazon	Obtained	3	\$13.99	41.97
8XAA 12V battery holder case box	Co-Rode	Amazon	Obtained	1	\$8.00	8
PCS 65mm TT Motor Motor Wheels	ALLPARTZ	Amazon	Obtained	1	\$3.99	3.99
1000PC popsicle sticks 4.5"	Darice	Amazon	Obtained	1	\$21.99	21.99
5.5" bolt and nut	HomeDepot	Home Depot	Obtained	1	\$4.99	4.99
glue gun & accessories	Amazon	Amazon	Obtained	1	\$8.99	8.99
4" RUBBER TRACKS	Super Droid Robots	Super Droid Robots	Avail.	2	\$35.99	71.98
Passives	TBD	Digikey	-	-	-	47.78
PCB Boards	JLPCB	TBD	-	5	-	43.64
					Total	\$2,224.58

Figure 22. Breakdown of Costs

Cost of Labor

Engineer	Rate	Hours	Total
Rylee Bessler	\$85/hr	500	\$106,250
Luis Chavez	\$70/hr	500	\$87,500
Ricardo Gallegos	\$100/hr	500	\$125,000
Minhhanh Tran	\$100/hr	500	\$125,000
Benson To	\$80/hr	500	\$100,000
Total Cost of Labour		\$543,760	
Total Cost of Parts		2224.58	
GRAND TOTAL = LABOR + PARTS		\$545,984.58	

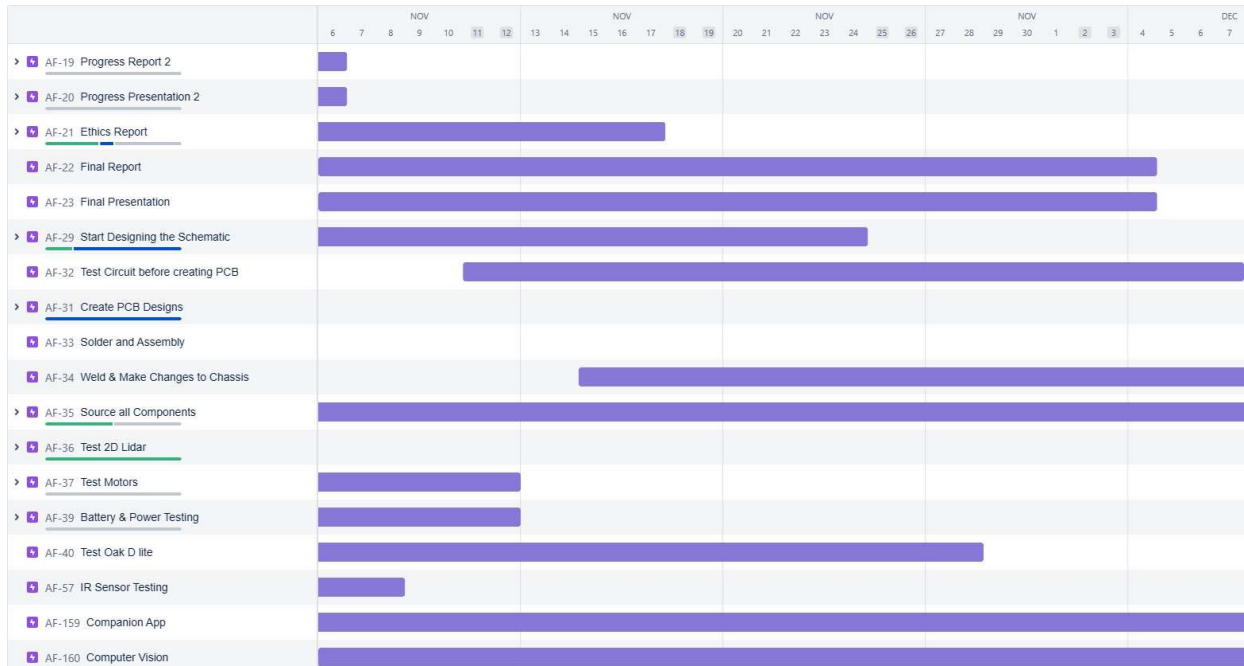
Schedule**Planned Timeline:**

Figure 23. Fall 2023 Schedule

CECS-490B: Roadies



Figure 24. Spring 2024 Schedule

Actual Timeline (sans coursework):

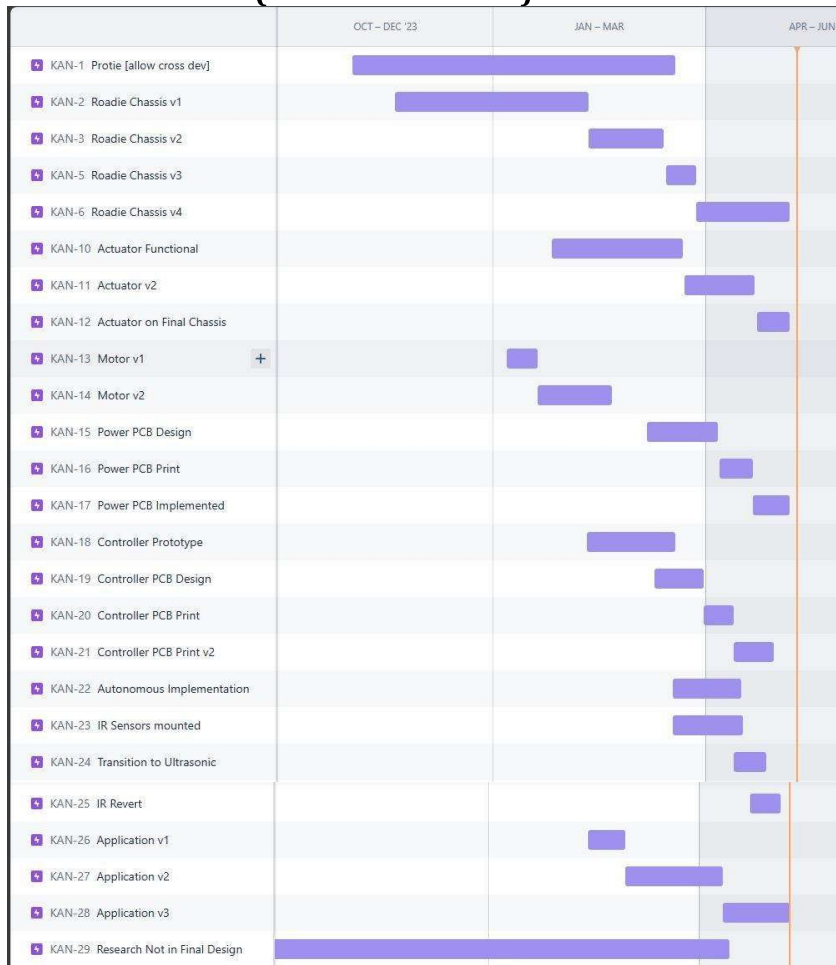


Figure 25. Actual Timeline

When faced with challenges, our team demonstrated remarkable creativity and unity. We came together to reevaluate our designs, brainstorming innovative solutions and making the tough decisions necessary for progress. Each member brought their unique perspective to the table, fostering an environment of collaboration and open communication. Through our collective efforts, we were able to quickly adjust in the moment and make progress towards our final vision. We realize that we had underestimated the amount of work necessary to accomplish all our goals within the given time. However, we remained focused, and continued to work collaboratively to meet our goals and are happy with our results.

Stretch Goals

A couple of stretch goals that we have if we were given more time to complete this project after acquiring all the knowledge that we have now obtained after finishing would be to change the motors and battery from a 12V source to a 24V and go back to using the tread tracks that were used before. The reason for this is that it will allow us to accomplish the sight that we had before which was to include a stairclimbing feature on the robot using the tread tracks but that was not possible due to an error on our end where the vibrations from moving caused us issues. If we had been able to mitigate this issue and change our motors and power supply, we believe that we would be even more satisfied with our project than it is right now. Finally, finish creating the companion app as we ran into issues implementing the Bluetooth as we did not have enough knowledge when we decided to create a companion app. We later found out how ambitious we were to try and include this but after further review this would be another stretch goal to keep in mind from where we finished off our project.

Outstanding Concerns and Expected Path if Continued

To enhance maneuverability of the Roadie, we upgraded the wheel size from 8 inches to 10 inches, a modification aimed at improving the suitability for potential future stair climbing capabilities. During the fabrication process, we had to revisit and modify half of the treads because the larger wheels initially did not clear one side. After some careful adjustments, the new 10-inch wheels seamlessly aligned with the bike chain, nuts, and bolts.

The prototype frame's weight was substantial due to maple wood. To address this weight concern while maintaining the structural integrity, we strategically drilled 1-inch holes every 2 inches along the frame. This design alteration ensured a lighter overall frame without compromising rigidity. During the stress testing, we observed that the wooden chassis, with the screws rigidly attached to the cross members, began to exhibit outward camber when subjected to a load exceeding 95 pounds. This weight limit was more than double the original specification. Due to this result, a solution would involve using a lighter and more durable chassis material such as aluminum.

Another problem was finding a way to track the user as the Oak-D Lite cannot select an object and just focus on that. So, an alternative that we produced was to track the user through Bluetooth and use a GPS module to be able to get the location of the Roadie to calculate the distance between the user and the Roadie. GPS proved to be an additional technology that we could not implement within the time allotment of Semester B and relied on IR to dictate when Roadie stopped and continued to move.

Project Ethical Concerns

Areas for potential ethical concerns:

- **Safety:** Both in the prototyping and in the use of the Roadie there are concerns around safety for all individuals involved. Naturally, our product relies on a battery source which if not properly accounted for can lead to overheating, fires, and exposure to chemicals. Additionally, we have multiple moving components which lend themselves to catching on extremities, so we intend to enclose as many systems as possible to minimize injury. While in manual mode, we need to make sure that the Roadie stops moving in case it loses connection to the controller so that there is no injury or property damage.
- **Environmental Impact:** All things at some point must return to the Earth. This is an area where our attention is lacking in our approach to the project. However, if we were to scale and take this idea to a commercial level our materials and components would need to be evaluated against their environmental impact. Furthermore, we could work to implement recycling programs or make it modular enough that hacking communities would be able to repurpose the subsystems of the Roadie.
- **Privacy and Security:** Now with the implementation of a companion app on the user's device there is an element of concern over data that we are pulling, especially if we were to assist the users in creating favorited or predefined paths. If this is ever offloaded to a cloud, we could be jeopardizing our user's safety. Additionally, as our project stands it has components that are very much IoT (Internet of Things) and could pose security risks when connected to the user's local network.
- **Right to Repair:** Needing to ensure that documentation is easily available to those that want to carry out their own repairs and that they have access to common tools and any software needed in the repair process.
- **Equity:** That this project if commercialized does not just benefit those that are in positions to purchase it while disproportionately having a negative impact on communities that would be sourcing many of the materials from the Earth that go into the manufacturing of Roadie.

Intellectual Property

What makes our project unique compared to what was already made, is how we are using more affordable parts compared to big companies like Amazon with their Amazon Scout. A design that uses both lidar and object detection like the Oak D Lite is a video by Scott's Robot Fun on YouTube in which he uses both hardware for his robot to follow him around his house through Nav2 Dynamic Object Following.

We also designed a frame from a 3d printer for testing our software development before the Roadie is completed. This custom design allows for components, power systems, and motors to be attached or placed on the Prodie (mini prototype) for easy access and debugging. The miniature scissor jack mechanism that goes on the Prodie was also custom designed on cad software tailored to our project. This allows us to save money on materials by simplifying the design.

Appendix

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Roadies GitHub Repository

<https://github.com/Luischavez02/ROADIES>